

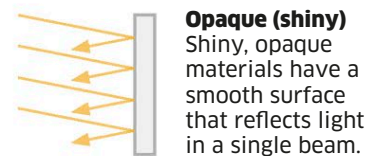
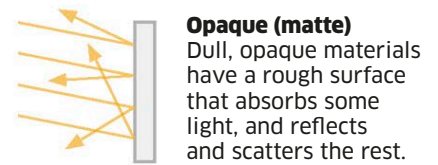
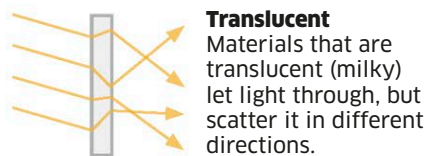
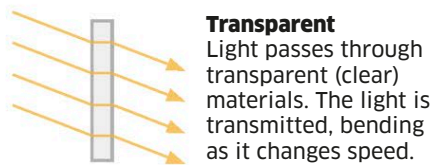
Light

Light is a type of electromagnetic radiation. It is carried by a stream of particles, called photons, that can also behave like a wave.

The most important source of light on Earth is the sun. Sunlight is produced by energy generated in the sun's core. Like the sun, some objects such as candles emit (send out) light—they are luminous. In contrast, most objects reflect and/or absorb light. Light travels as transverse waves, like ripples in water; the direction of wave vibration is at right angles to the direction that the light travels.

Light and matter

A material appears shiny, dull, or clear depending on whether it transmits, reflects, or absorbs light rays. Most materials absorb some light.



Sources of light

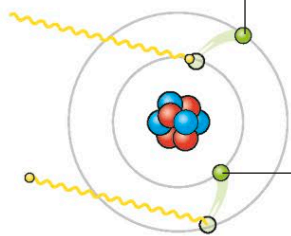
Light is a form of energy. It is produced by two distinct processes: incandescence and luminescence. Incandescence is the emission of light by hot objects. Luminescence is the emission of light without heat.

Photons

If an atom gains energy, electrons orbiting the nucleus jump to higher orbits, or "energy levels." When the electrons return to their original orbits, they release photons of light, or other electromagnetic radiation.

Excited atom

Electron jumps to higher level when it gains energy.

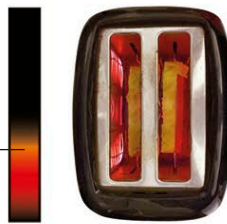


Incandescence

Incandescent light sources produce light because they are hot. The hotter an object, the more of the visible color spectrum it produces. Incandescent light produces all the colors in its range in a continuous spectrum.

Toaster grill

A grill or the element of a toaster, at about 1,110°F (600°C), will glow with only red light. It also emits light in the infrared range.



Color spectrum

A spectroscopic image shows the spectrum of colors a light source emits.

Candle flame

A candle flame, at about 1,550°F (850°C), produces some green and yellow light, as well as red, so it glows with a bright yellow light.



Incandescent light bulb

The filament of an old-style light bulb, at about 4,500°F (2,500°C), produces nearly all the spectrum. Missing some blue light, it has a yellow tinge.



Luminescence

A luminescent light source produces light by electrons losing energy in atoms. Energy is lost in exact amounts, which determine the color of the light produced, depending on the chemistry of the luminescent material.

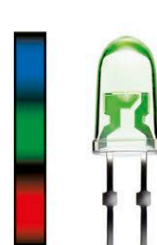
Bioluminescence

Bioluminescent animals such as fireflies produce a single wavelength of yellowish-green light by oxidizing a molecule called luciferin.



Light-emitting diode (LED)

An LED may produce two or more colors. Energy-saving LEDs produce red, green, and blue light, chosen to give an impression of white light.



Compact fluorescent lamp

Luminescent paints on the inside of glass produce red, green, and blue light, giving an impression of white light (not a continuous spectrum).



Lasers

A laser produces an intense beam of light of a single wavelength. The light is concentrated in a "lasing medium" such as crystal. In a crystal laser, light from a coiled tube "excites" atoms in a tube made of crystals, such as ruby. The photons of light that these excited atoms produce reflect between the tube's mirrored ends and escape as a powerful beam. We say the light is coherent, because the waves are in step.

